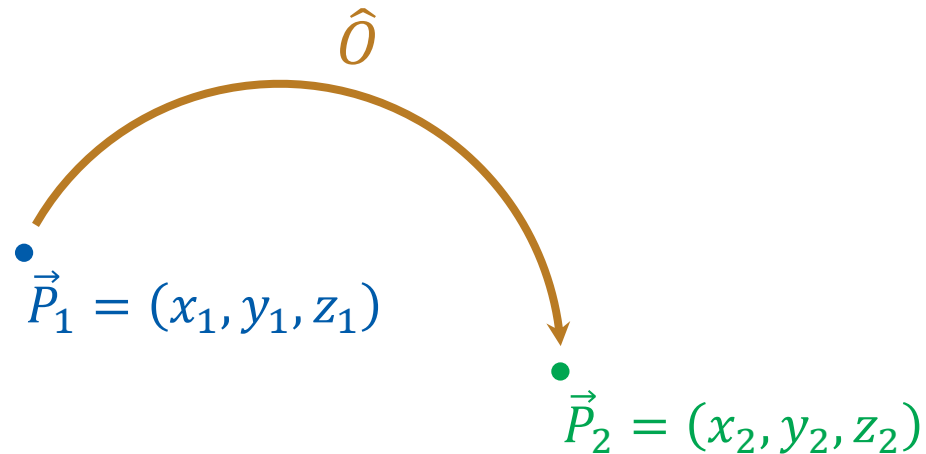


화학수학

열다섯 번째 수업

행렬

Symmetry Operators



$$\hat{O}(x_1, y_1, z_1) = (x_2, y_2, z_2)$$

$$\hat{O}\vec{P}_1 = \vec{P}_2$$

Point Symmetry Operators

(점대칭 연산자)

a class of symmetry operators
that leave **at least one point unmoved**
(the unmoved point = origin)

Identity operator $\hat{E}$

Inversion operator $\hat{i}$

Reflection operator $\hat{\sigma}$

Rotation operator $\hat{C}_n, \hat{S}_n$

Symmetry Elements (대칭 요소)

Point, line, or plane relative to which
the symmetry operation is performed

- Must include the origin
- Sometimes denoted by the same symbol
- Inversion i : the origin
- Reflection σ : the reflection plane
- Rotation C_n, S_n : the rotation axis

Symmetry of Object

- If after a symmetry operation, all particles in the object is in the same conformation as before, **the symmetry operator belongs to the object.**
- Coordinate system
 - origin: center of mass of the object
 - z axis: the rotation axis of highest order (largest n)

Symmetry Operators on Functions

If an operator $\hat{O}$ operates as $\hat{O}(x_1, y_1, z_1) = (x_2, y_2, z_2)$,
for a function f ,

$$\hat{O}f = g$$

where $g(x_2, y_2, z_2) = f(x_1, y_1, z_1)$.

Use the relation between (x_1, y_1, z_1) and (x_2, y_2, z_2) !

Eigenfunctions of Symmetry Operators

- Eigenvalue = +1

the operator leaves a function unchanged

- Eigenvalue = -1

the operator changes the sign of the function's value
at all points

오늘의 수업

- Matrices
 - Matrix and matrix algebra
 - Special matrices
 - Partners of a matrix
 - Properties of a matrix

Matrix (행렬)

A list of quantities

arranged in rows (행) and columns (열)

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}$$

matrix elements
(행렬 원소)

(m by n matrix)

Some Special Matrices

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}$$

- $m = n$: square matrix (정사각행렬)
- $m = 1$: row vector (행벡터)
- $n = 1$: column vector (열벡터)

More on Square Matrices (1)

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{bmatrix}$$

off-diagonal elements
(비대각원소)

diagonal elements
(대각원소)

More on Square Matrices (2)

- **Diagonal matrix** (대각행렬): off-diagonal elements = 0

$$\mathbf{A} = \begin{bmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{bmatrix}$$

- **Upper triangular matrix**
(위삼각행렬)

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a_{22} & a_{23} \\ 0 & 0 & a_{33} \end{bmatrix}$$

- **Lower triangular matrix**
(아래삼각행렬)

$$\mathbf{A} = \begin{bmatrix} a_{11} & 0 & 0 \\ a_{21} & a_{22} & 0 \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

Matrix Algebra

- Two matrices are **equal** if and only if (1) they have the same numbers of rows and columns and (2) every element of one matrix is equal to the corresponding element of the other matrix

- **Sum of two matrices:** $\mathbf{C} = \mathbf{A} + \mathbf{B}$ if and only if (iff)

$$c_{ij} = a_{ij} + b_{ij} \text{ for every } i \text{ and } j$$

- **Product of a scalar c and a matrix $\mathbf{A}$:** $\mathbf{B} = c\mathbf{A}$ iff

$$b_{ij} = ca_{ij} \text{ for every } i \text{ and } j$$

Product of Two Matrices

- Matrix product $\mathbf{C} = \mathbf{AB}$ iff

$$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$$

where n is the number of columns in $\mathbf{A}$ and the number of rows in $\mathbf{B}$.

$$(m \text{ by } n \text{ matrix}) \times (n \text{ by } p \text{ matrix}) = (m \text{ by } p \text{ matrix})$$

Example

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \times \begin{bmatrix} 10 & 11 \\ 20 & 21 \\ 30 & 31 \end{bmatrix}$$

$c_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$

$$= \begin{bmatrix} 1 \times 10 + 2 \times 20 + 3 \times 30 & 1 \times 11 + 2 \times 21 + 3 \times 31 \\ 4 \times 10 + 5 \times 20 + 6 \times 30 & 4 \times 11 + 5 \times 21 + 6 \times 31 \end{bmatrix}$$

$$= \begin{bmatrix} 10 + 40 + 90 & 11 + 42 + 93 \\ 40 + 100 + 180 & 44 + 105 + 186 \end{bmatrix} = \begin{bmatrix} 140 & 146 \\ 320 & 335 \end{bmatrix}$$

Example 13.15

Find the matrix product

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & -1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & 0 & 2 \\ 3 & 0 & 1 \\ 1 & 1 & -1 \end{bmatrix}.$$

The matrix product is

$$\begin{bmatrix} 1 \times 0 + 0 \times 3 + 2 \times 1 & 1 \times 0 + 0 \times 0 + 2 \times 1 & 1 \times 2 + 0 \times 1 + 2 \times (-1) \\ 0 \times 0 + (-1) \times 3 + 1 \times 1 & 0 \times 0 + (-1) \times 0 + 1 \times 1 & 0 \times 2 + (-1) \times 1 + 1 \times (-1) \\ 0 \times 0 + 0 \times 3 + 1 \times 1 & 0 \times 0 + 0 \times 0 + 1 \times 1 & 0 \times 2 + 0 \times 1 + 1 \times (-1) \end{bmatrix}$$

which is

$$\begin{bmatrix} 2 & 2 & 0 \\ -2 & 1 & -2 \\ 1 & 1 & -1 \end{bmatrix}$$

Scalar Product of Vectors

For two vectors $\vec{F} = (F_x, F_y, F_z)$ and $\vec{G} = (G_x, G_y, G_z)$, their scalar product can be seen as

$$\vec{F} \cdot \vec{G} = F_x G_x + F_y G_y + F_z G_z = \begin{bmatrix} F_x & F_y & F_z \end{bmatrix} \begin{bmatrix} G_x \\ G_y \\ G_z \end{bmatrix}$$

- We typically represent vectors as **column vectors**

Matrix as Vector Operator

A vector $\vec{v}$ can be multiplied by a square matrix $\mathbf{A}$:

$$\vec{u} = \mathbf{A}\vec{v}$$

$$\begin{bmatrix} u_x \\ u_y \\ u_z \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}$$

This can be seen as

$$\vec{v} \xrightarrow{\mathbf{A}} \vec{u}$$

Properties of Matrix Multiplication

- Associative (결합법칙)

$$\mathbf{A}(\mathbf{BC}) = (\mathbf{AB})\mathbf{C}$$

- Distributive (분배법칙)

$$\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$$

- Not necessarily commutative (교환법칙 X)

$$\mathbf{AB} \neq \mathbf{BA} \text{ (in some cases)}$$

What's Next: Full Bag of New Terms

1. **Special matrices:** identity matrix, zero matrix
2. **Partner of a matrix:** inverse, transpose, Hermitian conjugate
3. **Properties of a matrix:** singular, symmetric, Hermitian, orthogonal, unitary
4. **Derived quantities from a matrix:** trace, determinant

Identity Matrix (항등행렬) **E**

$$\mathbf{EA} = \mathbf{AE} = \mathbf{A}$$

$$\mathbf{E} = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

$$E_{ij} = \delta_{ij} = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases}$$

Zero Matrix (영행렬) $\mathbf{0}$

$$\mathbf{0A} = \mathbf{A0} = \mathbf{0}$$

$$\mathbf{0} = \begin{bmatrix} 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix}$$

$$0_{ij} = 0$$

Inverse of Matrix (역행렬)

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{E}$$

- Only square matrices have inverses
- In terms of matrix elements,

$$\sum_{k=1}^n a_{ik}(A^{-1})_{kj} = \delta_{ij}$$

Finding Inverse: Gauss-Jordan Elimination

$$\mathbf{A}\mathbf{A}^{-1} = \mathbf{E}$$

- Apply a series of **row operations** on $\mathbf{A}$ and $\mathbf{E}$ to obtain

$$\mathbf{E}\mathbf{A}^{-1} = \mathbf{D}$$

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} (A^{-1})_{11} & (A^{-1})_{12} \\ (A^{-1})_{21} & (A^{-1})_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} (A^{-1})_{11} & (A^{-1})_{12} \\ (A^{-1})_{21} & (A^{-1})_{22} \end{bmatrix} = \begin{bmatrix} d_{11} & d_{12} \\ d_{21} & d_{22} \end{bmatrix}$$


Row Operations

- Scalar Multiplication

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \rightarrow \begin{bmatrix} ca_{11} & ca_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

- Row subtraction

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \rightarrow \begin{bmatrix} a_{11} - a_{21} & a_{12} - a_{22} \\ a_{21} & a_{22} \end{bmatrix}$$

- Combination of the two

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \rightarrow \begin{bmatrix} a_{11} - ca_{21} & a_{12} - ca_{22} \\ a_{21} & a_{22} \end{bmatrix}$$

Gauss-Jordan Elimination

- Construct an augmented matrix

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} (A^{-1})_{11} & (A^{-1})_{12} \\ (A^{-1})_{21} & (A^{-1})_{22} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \rightarrow \left[\begin{array}{cc|cc} a_{11} & a_{12} & 1 & 0 \\ a_{21} & a_{22} & 0 & 1 \end{array} \right]$$

- Make an upper triangular matrix with diagonals = 1

$$\left[\begin{array}{cc|cc} 1 & * & * & * \\ 0 & 1 & * & * \end{array} \right]$$

- Then make the identity matrix

$$\left[\begin{array}{cc|cc} 1 & 0 & * & * \\ 0 & 1 & * & * \end{array} \right] A^{-1}$$

Example 13.16

Find the inverse of the matrix

$$\mathbf{A} = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}.$$

Construct an augmented matrix:

$$[\mathbf{A}|\mathbf{E}] = \left[\begin{array}{ccc|ccc} 2 & 1 & 0 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array} \right]$$

Step 1: upper triangular matrix

Step 2: identity matrix

Example 13.16

$$\left[\begin{array}{ccc|ccc} 2 & 1 & 0 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array} \right]$$

(1) Making upper triangular matrix

(Row 1) $\times 1/2$:

$$\left[\begin{array}{ccc|ccc} 1 & 1/2 & 0 & 1/2 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array} \right]$$

(Row 2) $-$ (Row 1):

$$\left[\begin{array}{ccc|ccc} 1 & 1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 3/2 & 1 & -1/2 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array} \right]$$

Example 13.16

$$\left[\begin{array}{ccc|ccc} 1 & 1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 3/2 & 1 & -1/2 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array} \right]$$

(Row 2) $\times$ (2/3):

$$\left[\begin{array}{ccc|ccc} 1 & 1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 1 & 2/3 & -1/3 & 2/3 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array} \right]$$

(Row 3) $-$ (Row 2):

$$\left[\begin{array}{ccc|ccc} 1 & 1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 1 & 2/3 & -1/3 & 2/3 & 0 \\ 0 & 0 & 4/3 & 1/3 & -2/3 & 1 \end{array} \right]$$

Example 13.16

$$\left[\begin{array}{ccc|ccc} 1 & 1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 1 & 2/3 & -1/3 & 2/3 & 0 \\ 0 & 0 & 4/3 & 1/3 & -2/3 & 1 \end{array} \right]$$

(Row 3) $\times$ (3/4):

$$\left[\begin{array}{ccc|ccc} 1 & 1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 1 & 2/3 & -1/3 & 2/3 & 0 \\ 0 & 0 & 1 & 1/4 & -1/2 & 3/4 \end{array} \right]$$

Now we have an upper triangular matrix with diagonals = 1.

Example 13.16

$$\left[\begin{array}{ccc|ccc} 1 & 1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 1 & 2/3 & -1/3 & 2/3 & 0 \\ 0 & 0 & 1 & 1/4 & -1/2 & 3/4 \end{array} \right]$$

(2) Making identity matrix

(Row 2) – (2/3) × (Row 3):

$$\left[\begin{array}{ccc|ccc} 1 & 1/2 & 0 & 1/2 & 0 & 0 \\ 0 & 1 & 0 & -1/2 & 1 & -1/2 \\ 0 & 0 & 1 & 1/4 & -1/2 & 3/4 \end{array} \right]$$

(Row 1) – (1/2) × (Row 2):

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3/4 & -1/2 & 1/4 \\ 0 & 1 & 0 & -1/2 & 1 & -1/2 \\ 0 & 0 & 1 & 1/4 & -1/2 & 3/4 \end{array} \right]$$

Example 13.16

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 3/4 & -1/2 & 1/4 \\ 0 & 1 & 0 & -1/2 & 1 & -1/2 \\ 0 & 0 & 1 & 1/4 & -1/2 & 3/4 \end{array} \right]$$

Now, the inverse is

$$\mathbf{A}^{-1} = \begin{bmatrix} 3/4 & -1/2 & 1/4 \\ -1/2 & 1 & -1/2 \\ 1/4 & -1/2 & 3/4 \end{bmatrix}$$

Check

$$\mathbf{A}\mathbf{A}^{-1} = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 3/4 & -1/2 & 1/4 \\ -1/2 & 1 & -1/2 \\ 1/4 & -1/2 & 3/4 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Transpose of Matrix (전치행렬) $\tilde{\mathbf{A}}$

Rows $\rightarrow$ columns, columns $\rightarrow$ rows

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}$$

$$\tilde{\mathbf{A}} = \begin{bmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \end{bmatrix}$$

$$(\tilde{\mathbf{A}})_{ij} = \tilde{a}_{ij} = a_{ji}$$

Scalar Product Revisited

For two column vectors

$$\mathbf{V} = \begin{bmatrix} v_x \\ v_y \\ v_z \end{bmatrix}, \quad \mathbf{U} = \begin{bmatrix} u_x \\ u_y \\ u_z \end{bmatrix},$$

their scalar product can be seen as

$$\mathbf{V} \cdot \mathbf{U} = v_x u_x + v_y u_y + v_z u_z = \begin{bmatrix} v_x & v_y & v_z \end{bmatrix} \begin{bmatrix} u_x \\ u_y \\ u_z \end{bmatrix} = \tilde{\mathbf{V}} \mathbf{U}$$

Hermitian Conjugate of Matrix $\mathbf{A}^\dagger$

(에르미트 켤레 행렬)

Transpose + complex conjugate of each element

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{bmatrix}$$

$$\mathbf{A}^\dagger = \begin{bmatrix} a_{11}^* & a_{21}^* & a_{31}^* \\ a_{12}^* & a_{22}^* & a_{32}^* \end{bmatrix}$$

$$(\mathbf{A}^\dagger)_{ij} = a_{ji}^*$$

Properties of a Square Matrix

- Singular matrix (특이행렬) has no inverse
- Symmetric matrix (대칭행렬): $\mathbf{A} = \tilde{\mathbf{A}}$
- Hermitian matrix (에르미트행렬): $\mathbf{A} = \mathbf{A}^\dagger$
- Orthogonal matrix (직교행렬): $\mathbf{A}^{-1} = \tilde{\mathbf{A}}$
- Unitary matrix (유니타리행렬): $\mathbf{A}^{-1} = \mathbf{A}^\dagger$

Trace of a Square Matrix (대각합)

$$\text{tr } \mathbf{A} = \sum_{i=1}^n a_{ii}$$

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \rightarrow \text{tr } \mathbf{A} = a_{11} + a_{22} + a_{33}$$

Property of Trace

Even when $\mathbf{AB} \neq \mathbf{BA}$,

$$\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$$

$$\begin{aligned}\text{tr}(\mathbf{AB}) &= \sum_{i=1}^n (\mathbf{AB})_{ii} = \sum_{i=1}^n \sum_{k=1}^n a_{ik} b_{ki} \\ &= \sum_{k=1}^n \sum_{i=1}^n b_{ki} a_{ik} = \sum_{k=1}^n (\mathbf{BA})_{kk} = \text{tr}(\mathbf{BA})\end{aligned}$$